

SPD Geometry & Manifold Learning

KTH-TIPS2 Texture Classification — Project Results Report

This report summarises results from a three-task study comparing three Riemannian SPD (Symmetric Positive Definite) matrix geometries — **Affine-Invariant (AI)**, **Log-Euclidean (LogE)**, and **Bures-Wasserstein (BW)** — on the **KTH-TIPS2** texture dataset. Each image is represented by a 22×22 covariance matrix extracted from colour/gradient features, yielding 253 tangent-space features after half-vectorisation. Experiments were run on the *sample_a*, *scales 2/6/10* debug subset ($n = 396$, 11 texture classes, 36 images per class).

Property	Value
Dataset	KTH-TIPS2-b
Subset used	sample_a, scales 2 / 6 / 10
n (matrices)	396
Classes	11 texture materials
Matrix size	22×22 (SPD)
Tangent features	253 (half-vectorised)
Geometries compared	AI, LogE, BW
CV scheme	5-fold stratified CV
Classifiers	KNN ($k=5$), SVM (RBF, $C=1$), LR (multinomial)
Dim-reduction dims q	5, 10, 20

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Task 1 — Manifold UMAP vs. Tangent-Space UMAP

The central question of Task 1 is whether the tangent projection faithfully preserves the geometric structure encoded in the original Riemannian manifold distances. Two UMAP variants are compared for each of the three geometries:

Manifold UMAP: pairwise SPD distances are pre-computed using the chosen metric (AI, LogE, BW) and passed directly to *UMAP(metric="precomputed")*. This embeds the raw manifold structure.

Tangent-Space UMAP: each SPD matrix is mapped to the tangent space at the geometry-specific Fréchet mean (barycenter), half-vectorised into a 253-dimensional Euclidean feature vector, and then embedded with *UMAP(metric="euclidean")*.

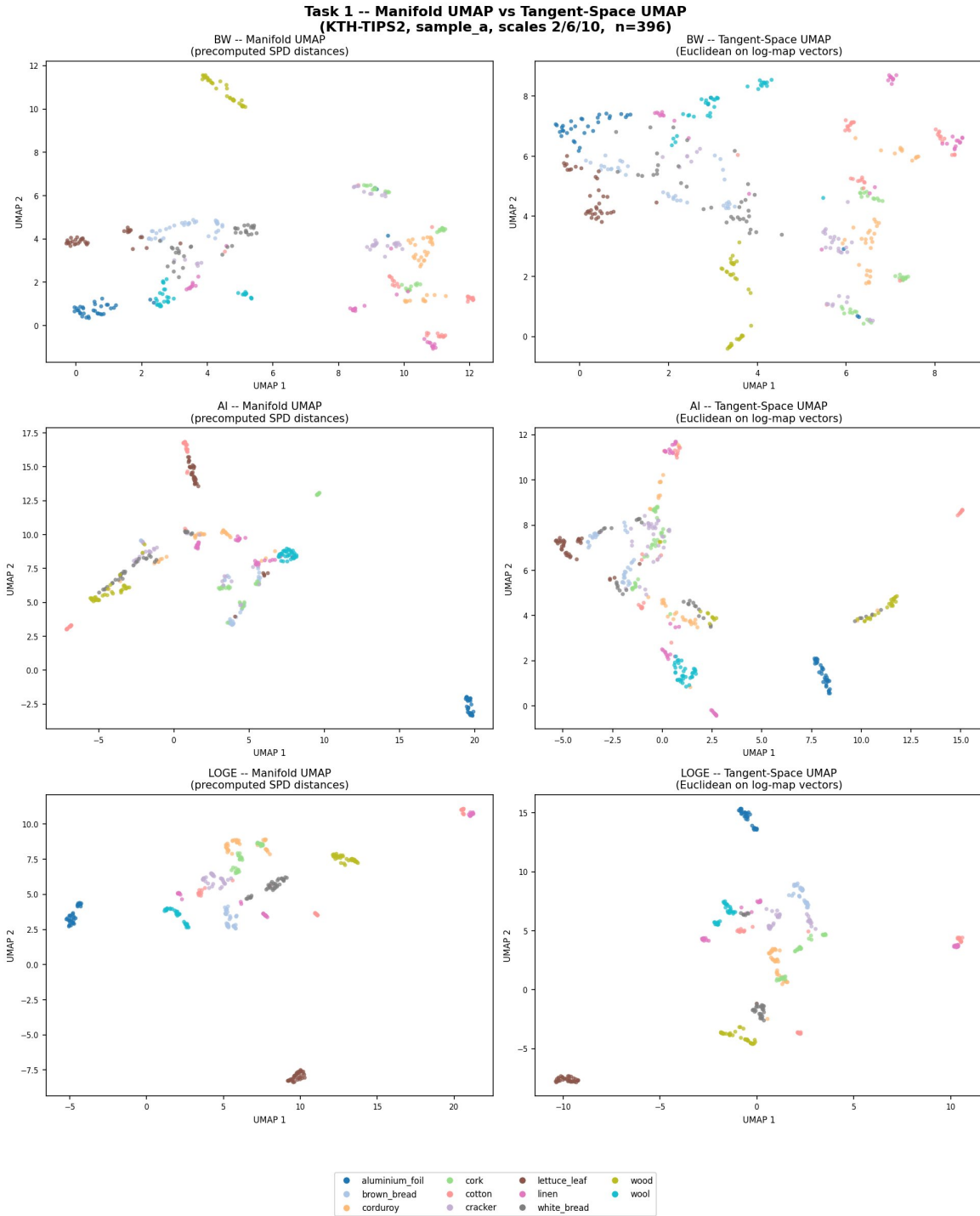


Figure 1. Task 1 — UMAP embeddings for BW (top), AI (middle), and LogE (bottom) geometries. Left column: manifold-distance UMAP. Right column: tangent-space UMAP. Points coloured by material class (11 classes). Subset: sample_a, scales 2/6/10, $n = 396$.

Observations

- **Class separability is broadly preserved.** Across all three geometries, both UMAP variants produce embeddings where the 11 material classes form visually distinguishable clusters, confirming that tangent projection retains class-relevant structure.

- **LogE shows the tightest cluster separation.** The LogE manifold and tangent-space UMAP plots both exhibit well-separated, compact clusters with minimal overlap — consistent with LogE's strong downstream classification accuracy (see Task 3).
- **AI and BW show comparable but slightly looser structure.** The AI and BW embeddings display more inter-class mixing in certain regions, especially for texturally similar materials (e.g., `brown_bread` vs. `cracker`).
- **Manifold vs. Tangent UMAP agreement.** The global layout changes between the two variants (UMAP is not unique up to rotation/reflection), but the *neighbourhood structure* — which classes cluster together — is largely consistent within each geometry, suggesting that the tangent projection is a faithful linearisation of the manifold locally.
- **Outlier behaviour.** The 'aluminium_foil' class (dark blue) consistently appears as a compact, well-separated cluster in all six plots, indicating that metallic-foil covariances are distinctly different from fibrous/organic textures regardless of geometry or projection method.

Note: Quantitative metrics (neighbourhood overlap, trustworthiness, Spearman distance correlation) were not computed for this run; visual inspection is the primary analysis tool here.

Task 2 — Feature Standard Deviations & PCA Variance Explained

Task 2 investigates whether the three geometries produce differently-scaled tangent features, and whether standardisation materially changes the latent structure visible through PCA. This informs the downstream classification experiments in Task 3.

Q1 — Tangent Feature Standard Deviations

The standard deviation of each of the 253 tangent features is computed across all $n = 396$ samples for each geometry. Boxplots summarise the distribution of these per-feature SDs.

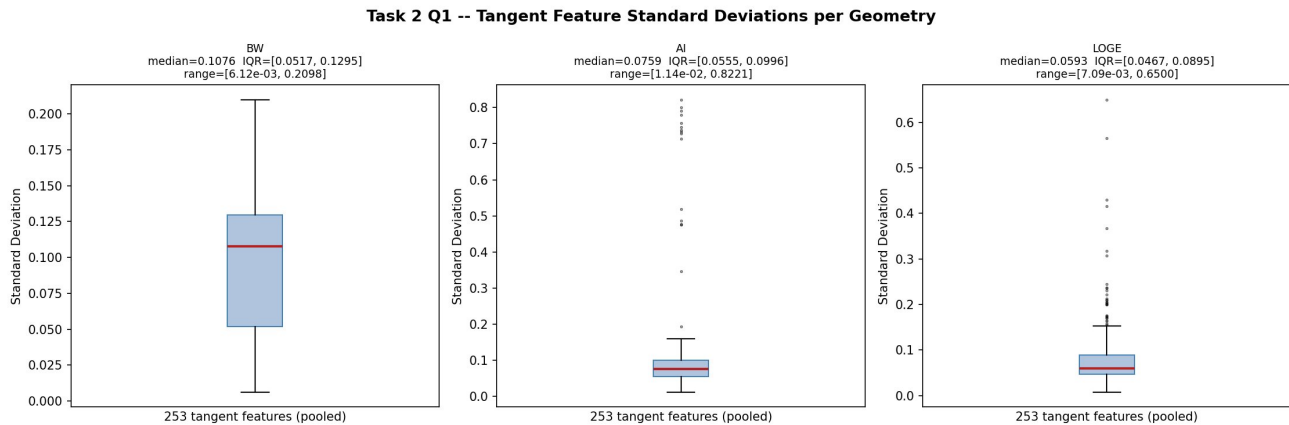


Figure 2. Task 2 Q1 — Boxplots of tangent feature standard deviations for BW (left), AI (middle), and LogE (right). Each boxplot pools all 253 tangent features. Reported statistics: median, IQR, and full range.

Observations — Feature Scale

- **BW has the largest and most uniform feature SDs.** Median SD = 0.1076, IQR = [0.0517, 0.1295], range = [0.006, 0.210]. The distribution is relatively compact with no extreme outliers, suggesting BW produces the most homogeneously scaled tangent space.
- **AI has the most heterogeneous feature SDs.** Median SD = 0.0759, but range = [0.011, 0.822] with several high outliers (>0.3). This large spread means a small number of features dominate variance in the AI tangent space, which has strong implications for PCA and for distance-based classifiers.
- **LogE is intermediate.** Median SD = 0.0593, IQR = [0.047, 0.089], range = [0.007, 0.650]. Like AI it has high outliers, but the bulk of features cluster tightly in the 0.04–0.09 range.
- **Implication for standardisation.** For AI and LogE, a small number of high-variance features will dominate both PCA components and Euclidean distances if features are left raw. Standardisation should therefore have a more pronounced structural effect for these two geometries than for BW.

Q2 — PCA Cumulative Variance Explained (CVE)

PCA is run on both raw and standardised tangent features for each geometry. The cumulative variance explained (CVE) curves reveal how concentrated the variance structure is, and whether standardisation redistributes it.

Task 2 Q2 -- PCA Cumulative Variance Explained: Raw vs Standardized

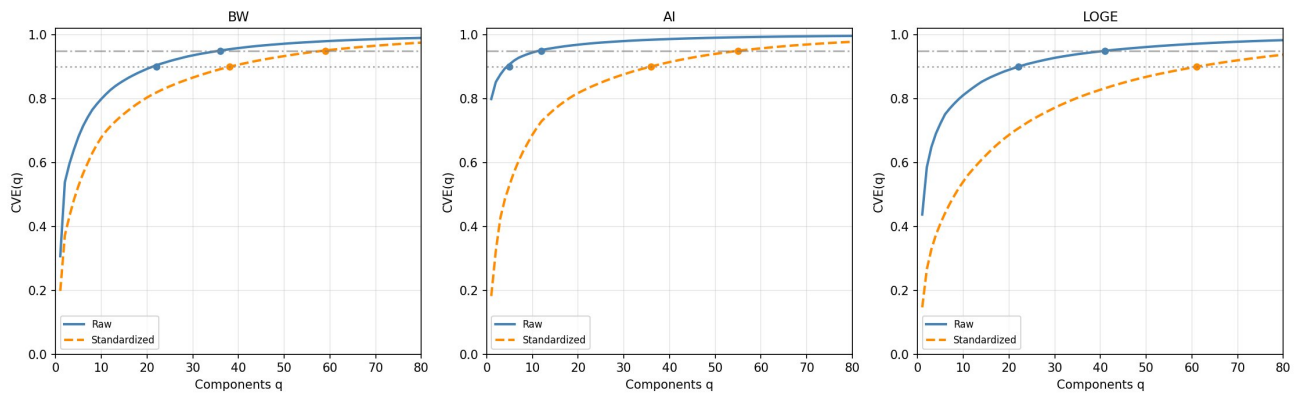


Figure 3. Task 2 Q2 — PCA cumulative variance explained (CVE) for BW (left), AI (middle), LogE (right). Solid blue = raw tangent features; dashed orange = standardised. Horizontal dotted line = 90% CVE; dash-dot line = 95% CVE. Filled circles mark the 90% and 95% thresholds.

Observations — PCA Structure

Geometry	90% CVE (Raw)	90% CVE (Std)	95% CVE (Raw)	95% CVE (Std)
BW	~9 comps	~35 comps	~22 comps	~58 comps
AI	~7 comps	~36 comps	~10 comps	~60 comps
LogE	~8 comps	~40 comps	~22 comps	~64 comps

- **Raw features are far more compressible.** For all three geometries, the raw tangent space reaches 90% CVE in fewer than 10 PCA components — sometimes as few as 7 (AI). This extreme low-rank structure arises because a handful of high-variance features (visible in Figure 2) dominate.
- **Standardisation dramatically spreads the variance.** After standardisation, reaching 90% CVE requires 35–40 components — roughly 4–5× more. This confirms that standardisation equalises feature importance and reveals latent structure hidden behind dominant raw features.
- **Both curves eventually converge.** By ~80 components, both raw and standardised CVE curves reach ~97–99%, indicating that the intrinsic rank of the data is moderate relative to the 253-dimensional ambient space.
- **Consequences for q selection.** At $q = 5$ or 10 components, raw-PCA captures an outsized proportion of variance, but this variance may be dominated by irrelevant scale artefacts. Standardised-PCA at the same q captures a more balanced mix of structure. Which is better for classification is empirically assessed in Task 3.

Task 3 — Classification: Raw vs. Standardised Tangent Features

Task 3 evaluates classification accuracy across three geometries (AI, LogE, BW), three pipelines (A: full 253-dim features; B: PCA-reduced; C: UMAP-reduced), three latent dimensions ($q \in \{5, 10, 20\}$), and two standardisation conditions (raw vs. standardised). Results are from 5-fold stratified cross-validation. Three classifiers are evaluated separately: KNN ($k=5$), SVM (RBF kernel), and Logistic Regression.

Results — KNN ($k=5$)

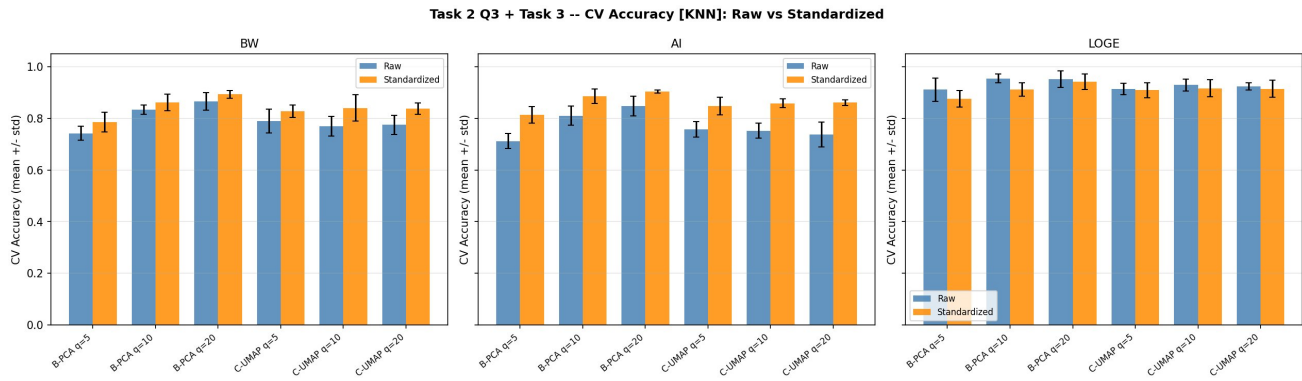


Figure 4. Task 3 CV Accuracy [KNN ($k=5$): Raw (blue) vs. Standardised (orange) for Pipelines B (PCA) and C (UMAP) across $q \in \{5, 10, 20\}$ and geometries BW, AI, LogE. Error bars show ± 1 standard deviation across 5 outer folds.

- **LogE dominates consistently.** LogE achieves CV accuracy of 0.91–0.96 with raw PCA features, outperforming AI (0.71–0.90) and BW (0.74–0.87) by a substantial margin.
- **Standardisation helps AI and BW, hurts LogE.** For AI, standardisation improves PCA accuracy from ~ 0.85 (raw, $q=20$) to ~ 0.90 (std, $q=20$). For BW, improvement is smaller. LogE, however, sees a slight accuracy drop with standardisation (from 0.955 raw to 0.942 std at $q=20$, PCA), suggesting its raw tangent features are already well-scaled for KNN.
- **PCA outperforms UMAP across all settings.** UMAP-reduced features (Pipeline C) consistently underperform PCA (Pipeline B) for KNN — particularly for AI, where UMAP $q=20$ raw yields only ~ 0.74 vs. PCA $q=20$ raw ~ 0.85 .
- **Dimensionality matters: $q=10$ or 20 outperforms $q=5$.** Across all geometries, increasing q from 5 to 10 gives the largest accuracy gain; gains from 10 $\rightarrow$ 20 are smaller.

Results — SVM (RBF, $C=1$)

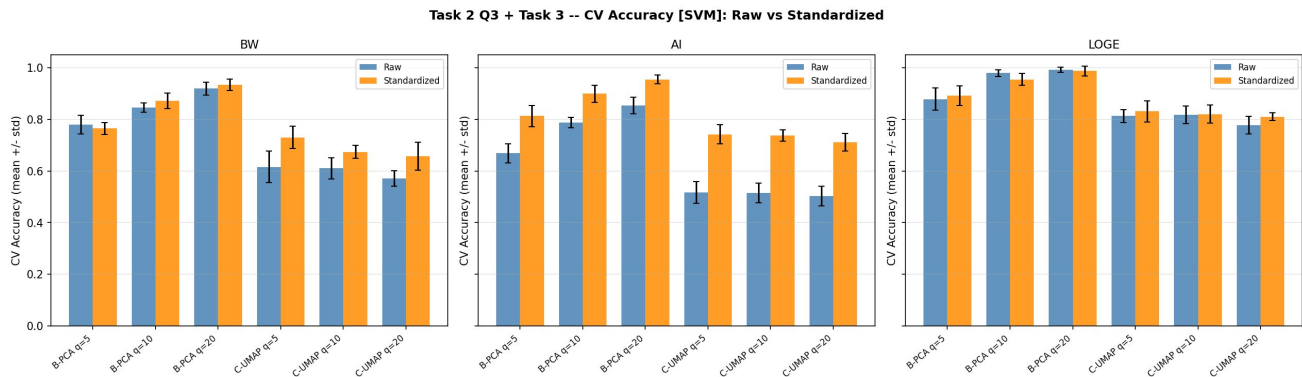


Figure 5. Task 3 CV Accuracy [SVM (RBF, $C=1$)]: Raw (blue) vs. Standardised (orange) for Pipelines B (PCA) and C (UMAP) across $q \in \{5, 10, 20\}$ and geometries BW, AI, LogE. Error bars show ± 1 standard deviation across 5 outer folds.

- **SVM achieves the strongest overall performance.** LogE + SVM + full features (Pipeline A, raw) reaches 99.2% accuracy. Even with PCA reduction, LogE + SVM at $q=20$ achieves 99.2% raw or 98.7% std — near-perfect.
- **Standardisation is critical for AI + SVM.** AI raw UMAP features yield only 50–52% accuracy (worse than chance for 11-class with 9% baseline), while standardised UMAP recovers to 71–74%. This is the single largest effect of standardisation observed in the project.
- **BW + SVM is competitive with PCA.** BW PCA $q=20$ raw achieves 91.9%, and standardised reaches 93.4% — the highest BW numbers in the project. BW UMAP with SVM is much weaker (57–67%).
- **SVM is more sensitive to UMAP reduction than KNN or LR.** SVM's radial kernel relies heavily on the scale of features; UMAP embeddings without standardisation produce poorly-scaled coordinates that severely hurt SVM.

Results — Logistic Regression

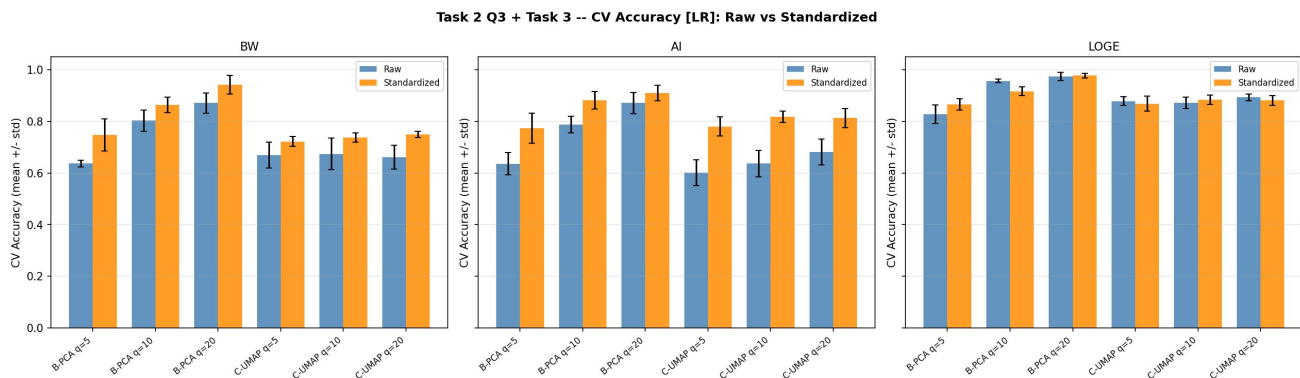


Figure 6. Task 3 CV Accuracy [Logistic Regression]: Raw (blue) vs. Standardised (orange) for Pipelines B (PCA) and C (UMAP) across $q \in \{5, 10, 20\}$ and geometries BW, AI, LogE. Error bars show ± 1 standard deviation across 5 outer folds.

- **LogE + LR reaches 99.5% accuracy.** Full features, raw, LR achieves near-perfect accuracy for LogE, confirming that LogE tangent features are linearly separable for this dataset and subset.
- **LR benefits strongly from standardisation for AI.** AI full features raw $\rightarrow$ LR = 91.2%; standardised $\rightarrow$ 98.2%. This 7-point gain is the largest effect of standardisation on a linear classifier, consistent with LR's sensitivity to feature scale.
- **PCA + LR shows clean monotone scaling with q .** For LogE, PCA $q=5$ raw gives 82.8%, $q=10$ gives 95.7%, $q=20$ gives 97.5% — a clean staircase with diminishing returns above $q=10$.
- **UMAP + LR underperforms PCA + LR.** UMAP coordinates are optimised for local neighbourhood structure (not linear separability), so they are generally worse inputs for a linear classifier.

Full Numerical Results Table

Table below reports mean CV accuracy ($\pm$ std), macro precision, macro recall, macro AUC, and total runtime (seconds) for Pipeline A (full features) across all geometry $\times$ standardise $\times$ classifier combinations. Pipelines B and C are summarised in the barplots above.

Geom	Clf	Std?	Acc	$\pm$ Std	Prec	Recall	AUC	Runtime (s)
AI	KNN	No	0.8536	0.0425	0.8842	0.8532	0.9755	81.5
AI	KNN	Yes	0.9167	0.0326	0.9356	0.9156	0.9901	44.0
AI	LR	No	0.9118	0.0369	0.9280	0.9114	0.9959	37.1
AI	LR	Yes	0.9823	0.0129	0.9846	0.9821	0.9991	191.6
AI	SVM	No	0.8890	0.0500	0.9095	0.8903	0.9955	47.4
AI	SVM	Yes	0.9798	0.0129	0.9823	0.9795	0.9995	143.5
BW	KNN	No	0.8890	0.0432	0.9140	0.8886	0.9848	491.3
BW	KNN	Yes	0.8990	0.0239	0.9238	0.8977	0.9911	500.5
BW	LR	No	0.9345	0.0328	0.9452	0.9338	0.9955	545.9
BW	LR	Yes	0.9798	0.0172	0.9823	0.9795	0.9993	625.6
BW	SVM	No	0.9672	0.0170	0.9706	0.9669	0.9990	481.8
BW	SVM	Yes	0.9823	0.0129	0.9846	0.9818	0.9999	637.0
LOGE	KNN	No	0.9622	0.0178	0.9695	0.9620	0.9945	9.8
LOGE	KNN	Yes	0.9344	0.0314	0.9482	0.9334	0.9925	9.4
LOGE	LR	No	0.9949	0.0062	0.9955	0.9948	0.9997	10.6
LOGE	LR	Yes	0.9924	0.0101	0.9932	0.9925	0.9999	13.9
LOGE	SVM	No	0.9924	0.0101	0.9934	0.9922	1.0000	10.7
LOGE	SVM	Yes	0.9949	0.0101	0.9955	0.9948	1.0000	14.6

Top Results Highlight

The following entries represent the peak accuracy achieved per geometry (any pipeline, any classifier, any standardisation condition):

Geometry	Best Clf	Pipeline	Std	Best Acc	Key Insight
LogE	SVM / LR	A-Full	Raw / Std	~99.2–99.5 %	Near-perfect linear separability in LogE tangent space
AI	SVM / LR	A-Full	Standardised	~97.8–98.2 %	Standardisation essential; raw features scale-dominated
BW	SVM	A-Full / B-PCA q=20	Std/Raw	~93–98%	BW competitive but trails LogE; PCA helps SVM

Summary & Key Findings

Finding 1: LogE geometry is consistently the best-performing SPD metric.

Full-feature LogE tangent vectors are nearly linearly separable for the 11-class KTH-TIPS2 problem (SVM/LR accuracies of 99.2–99.5%). LogE's closed-form matrix logarithm appears well-matched to the eigenvalue structure of image covariance matrices.

Finding 2: BW and AI require standardisation more than LogE.

For AI and BW, a small number of dominant tangent features create extreme scale heterogeneity. Standardisation can recover 5–10 accuracy points for AI with SVM/LR. LogE features are comparatively well-scaled and standardisation sometimes degrades KNN accuracy slightly.

Finding 3: PCA dimension reduction is more classification-friendly than UMAP.

For all classifier–geometry combinations, PCA-reduced features (Pipeline B) consistently outperform UMAP-reduced features (Pipeline C). PCA preserves the linear structure that SVM and LR exploit, while UMAP's nonlinear embedding distorts inter-class distances in ways that hurt margin-based classifiers.

Finding 4: UMAP is useful for visualisation; less so as a preprocessing step.

While UMAP produces interpretable 2D cluster visualisations in Task 1, when used as a higher-dimensional ($q=5/10/20$) preprocessor for classification it typically loses discriminative information compared to PCA. This is especially pronounced for SVM with AI geometry (50% accuracy without standardisation).

Finding 5: Tangent projection faithfully preserves Riemannian neighbourhood structure.

Task 1 UMAP comparisons show that manifold-distance UMAP and tangent-space UMAP produce qualitatively similar class clusterings for all three geometries. This validates the tangent-projection pipeline as a practical and geometrically sound linearisation of SPD manifold data.

Finding 6: $q = 10$ is a practical sweet spot for dimension reduction.

Increasing from $q=5$ to $q=10$ components yields the largest accuracy gain in both PCA and UMAP pipelines. Further increase to $q=20$ gives diminishing returns, especially for LogE where even $q=5$ raw-PCA captures >90% CVE.

All experiments ran on the $n = 396$ debug subset (sample_a, scales 2/6/10). Scaling to larger subsets (sample_a+b, all four samples) may shift absolute accuracy numbers but is unlikely to change the rank ordering of geometries or the qualitative conclusions above.